

TABLE III
CONTINUED

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of events	118	118	118	118	118	118	118	118
Number of observations	554,931	554,931	554,931	554,931	554,931	554,931	554,931	554,931
Number of workers in the sample	2,652,792	358,086	384,498	274,812	1,504,643	156,634	315,397	349,749
Sector	Overall	Tradeable	Nontradeable	Construction	Other	Restaurants	Retail	Manufacturing

Notes. The table reports the effects of a minimum wage increase by industries based on the event study analysis (see [equation \(1\)](#)) exploiting 138 state-level minimum wage changes between 1992 and 2016. The table reports five-year averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs at and up to \$5 above it, employment, and wages for all sectors (column (1)), tradeable sectors (column (2)), nontradeable sectors (column (3)), construction (column (4)), other sectors (column (5)), restaurants (column (6)), retail (column (7)), and manufacturing industries (column (8)). Our classification of tradeable, nontradeable, construction, and other sectors follows [Mian and Sufi \(2014\)](#) (see [Online Appendix D](#) for the details). Regressions are weighted by state-quarter aggregated population. Robust standard errors in parentheses are clustered by state; significance levels are *0.10, **0.05, ***0.01.

The first two rows report the change in number of missing jobs below the new minimum wage (Δa), and excess jobs above the new minimum wage (Δb) relative to the pretreatment total employment. The third row, the percentage change in average wages in the affected bins, ($\% \Delta W$), is calculated using [equation \(2\)](#). The fourth row, percentage change in employment in the affected bins, is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{b - 1}$). The fifth row, employment elasticity with respect to the minimum wage, is calculated as $\frac{\Delta a + \Delta b}{\% \Delta W}$, whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{b - 1}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

the minimum wage varies a lot across industries. The minimum wage is highly binding in the restaurant sector with a missing jobs estimate of 10.1%, whereas it does not appear to be binding in the construction sector. The minimum wage is more binding in the nontradeable sector (6.6%) than in the tradeable sector (1.6%) or in the manufacturing sector (1.7%).

The effect of the minimum wage on employment also varies by sector. We find that the number of excess jobs at or above the minimum wage is smaller than the missing jobs in the tradeable sector, and so the employment effect is negative (-11.1% , std. err. 13.6%), but not statistically significant. Similarly, the point estimate in the manufacturing sector suggests that around 10.1% (std. err. 14.5%) of the jobs directly affected by the minimum wage are destroyed. The implied employment elasticity with respect to own wage is quite large in magnitude in both sectors (-1.910 in the tradeable sector and -1.385 in manufacturing), although the estimates are imprecise and statistically insignificant.

At the same time, we find no indication for negative disemployment effects in the nontradeable, restaurant, and retail sectors where most minimum wage workers are employed in the United States. The employment elasticity with respect to own wage in the nontradeable sector is positive (0.387 , std. err. 0.597), which is in stark contrast to the tradeable sector, where we find a large negative elasticity. [Harasztosi and Lindner \(forthcoming\)](#) find similar sectoral patterns in Hungary and argue, using revenue data, that the larger job losses for tradeables reflect a more elastic consumer demand in that sector.

3. By Pretreatment Employment Status. We consider the effect of the minimum wage separately on workers who were employed prior to the minimum wage increase (incumbent workers) and for new entrants into the labor market. We partition our sample of wage earners into incumbent workers and new entrants by exploiting the fact that the CPS interviews each respondent twice, exactly one year apart.²⁸ The partition limits our sample to

the 1992–2016 period, we first replicate our benchmark analysis using all industries for this restricted sample in [Table III](#), column (1). The estimated employment and wage effects on this restricted sample are similar to the full 1979–2016 sample.

28. All CPS respondents are interviewed for four months in the first interview period, then rotated out of the survey for eight months, and then rotated back into the survey for a final four months of interviews. In the fourth month of each

the 1980–2016 time period covering 137 eligible minimum wage-raising events and also restricts our time window to one year around the minimum wage increase rather than the five years in our baseline sample.

Figure IV shows the event study estimates for new entrants (Panel A) and incumbents (Panel B) for each k -dollar wage bin relative to the new minimum wage. For both subgroups, new minimum wages clearly bind, with significantly fewer jobs just below and significant more at the new minimum. This highlights that studies that restricts their sample to incumbent workers (e.g., Currie and Fallick 1996; Abowd et al. 2000; Clemens and Wither 2019) can only provide a partial characterization of the full effects of the minimum wage increase, since new entrants are also affected by the policy.

For both groups the excess jobs closely match the missing jobs (for incumbents $\Delta a = 1.3\%$ and $\Delta b = -1.2\%$ and for new entrants $\Delta a = 0.6\%$ and $\Delta b = -0.5\%$), so the net employment changes are approximately 0. The green and blue dashed lines (color version available online) show the running sums of employment changes up to the corresponding wage bin for each group. The lines show that in both cases there is little change in upper-tail employment. We note that if employers are replacing lower-skilled workers with higher-skilled ones, we should expect to see some reduction in jobs for previously employed workers, perhaps offset by high-skilled entrants; the lack of job loss for incumbents provides additional evidence against such labor-labor substitution. The affected wage increase for incumbents (9.5%, std. err. 2.0%) is significantly larger than it is for new entrants (1.9%, std. err. 1.3%) and some of these differences can be explained by the lack of spillover effects for the new entrants. In the next section we return to this issue.

III.B. Wage Spillovers

So far we have focused on the employment effects of the minimum wage. However, an equally important question is understanding the nature of the wage effects. In this section, we quantify the direct effect of the minimum wage and the indirect effect that comes from wage spillovers.

interview period (the “outgoing rotation group”), respondents are asked questions about wages. Online Appendix E explains how we match workers across rotation groups.

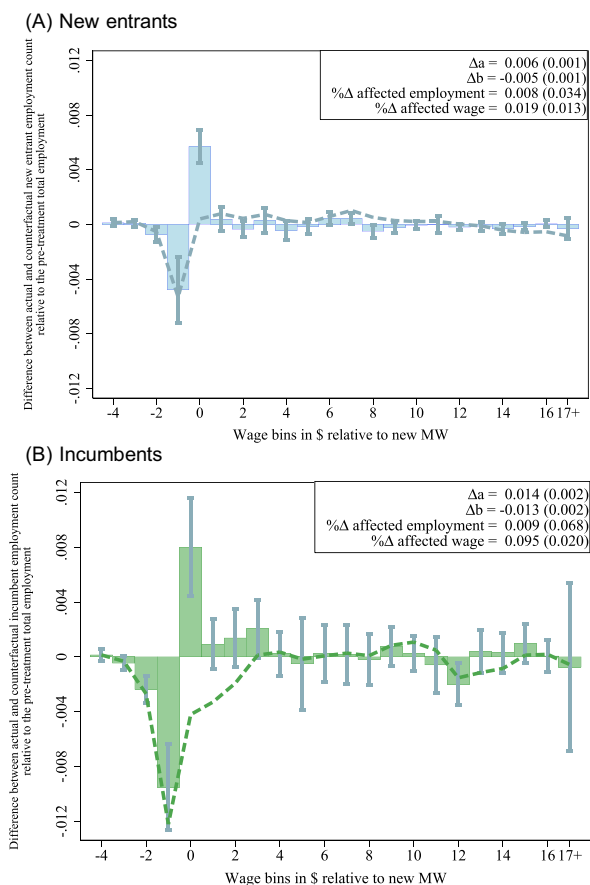


FIGURE IV

Impact of Minimum Wages on the Wage Distribution by Pretreatment Employment Status: New Entrants and Incumbents

The figure shows the main results for new entrants (Panel A) and for incumbents (Panel B) from our event study analysis (see [equation \(1\)](#)) exploiting 138 state-level minimum wage changes between 1979 and 2016. The blue bars in Panel A (color version available online) show for each dollar bin the estimated change in the number of new entrants in that bin one year post-treatment relative to the total employment of the new entrants one year before the treatment. The green bars in Panel B show the equivalent for incumbents. Incumbent workers were employed a year prior to the minimum wage increase, whereas new entrants were not. The error bars show the 95% confidence interval calculated using standard errors that are clustered at the state level. The dashed green and blue lines show the running sum of employment changes up to the wage bin they correspond to for new entrants and incumbents, respectively. The figures highlight that the ripple effect of the minimum wage mainly comes from incumbent workers.